

ICT 1404 Mathematics & Statistics for Computing
Assignment 01

1. (a) In how many ways can the letters of each of the following words be arranged?
(i) STATISTICS (ii) MATHEMATICS
(b) How many different words can be made using **four** letters of the following words?
(i) REENGINEERING (ii) MUALLABRALLUGUM
(c) How many different words can be made with the letters of the word MINIMUM which three letters MMM not with together?

- 2 The board of directors of a corporation comprises 10 members (6 are men and 4 are women). An executive board, formed by 5 directors, needs to be elected. How many possible ways are there to form the executive board in each of the following cases?:
 - a) Without any restrictions.
 - b) With at least 2 women
 - c) Must include a particular man and woman

- 3 Using matrices solve the following set of simultaneous equations.
 - a) $2x+y+3z=32$, $4x+3y+2z=41$, $3x+2y+5z=54$
 - b) $2x+2y+z=3$, $x+3y+3z=2$, $4x+3y+2z=4$
 - c) $3x-4y+5z = -6$, $x+ y-2z = -1$, $2x+3y+ z = 5$.

Submit on or before 14/02/ 2025